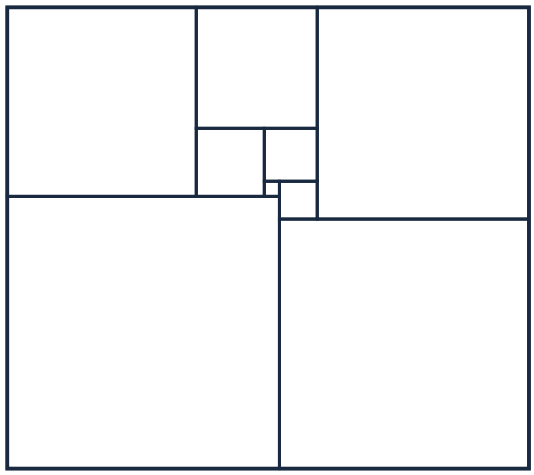
Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2000-I-Q04] | Part A (Q1-Q5)

Answer: 260



1. Label the central smallest square side x and the square immediately lower-right side y. Matching the segment lengths successively gives the other seven sides $x+y, 2x+y, 3x+2y, 5x+3y, 4x+4y, 4x+5y, 8x+4y$.

2. The two vertical boundaries have lengths $(8x+4y)+(5x+3y)$ and $(4x+5y)+(4x+4y)$. Equality gives $5x=2y$.

3. Write $x=2t, y=5t$. The rectangle dimensions are $69t, 61t$. As $\gcd(69,61)=1$, positive integral coprime dimensions force $t=1$. Hence perimeter $2(69+61)=260$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2004-II-Q05] | Part A (Q1-Q5)

Answer: 766

1. Normalize the scheduled time per quarter to one, so the full deadline is four units. The first three quarters use $1 + 1000/900 + 1000/800 = 121/36$ units.

2. The last quarter has $23/36$ time left and needs 1000 worker-time units. Thus the workforce must be at least $\lceil 36000/23 \rceil = 1566$.

3. Additional hires are $1566 - 800 = 766$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2006-I-Q05] | Part A (Q1-Q5)

Answer: 936

1. Square and match the independent radicals: $ab = 52, ac = 234, bc = 72$. Multiplying gives $(abc)^2 = 52 \cdot 234 \cdot 72 = 936^2$.

2. Positivity yields $abc = 936$. Indeed $(a, b, c) = (13, 4, 18)$ also gives constant part $338 + 48 + 1620 = 2006$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2007-I-Q02] | Part A (Q1-Q5)

Answer: 052

1. At time $t \geq 4$, positions are $6t, 10(t-2), 8(t-4)$. Set twice each candidate position equal to the sum of the others.

2. B as midpoint gives $t = 4/3$, inadmissible; C gives an inconsistent equation; A gives $12t = 18t - 52$, so $t = 26/3$. The midpoint position is $6t = 52$, before any finish.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2007-II-Q04] | Part A (Q1-Q5)

Answer: 450

1. Let one X,Y require u, v worker-hours. Equations $300u + 200v = 100$, $240u + 300v = 120$ give $u = 1/7, v = 2/7$.

2. Then $150/7 + 2m/7 = 150$, so $m = 450$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2008-I-Q01] | Part A (Q1-Q5)

Answer: 252

1. If the original total is N, the unchanged girl count gives $.60N = .58(N + 20)$, so $N = 580$.

2. The new dancing count is $.40 \cdot 580 + 20 = 252$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2008-II-Q02] | Part A (Q1-Q5)

Answer: 620

1. If R's speed is r miles/min, its time is $50/r + 49 \cdot 5$. J's time is $200/(3r) + 24 \cdot 5$.

2. Equating gives $125 = 50/(3r)$, hence $r = 2/15$. Total time $375 + 245 = 620$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2009-II-Q01] | Part A (Q1-Q5)

Answer: 114

1. If each stripe uses s, red use beyond its pure stripe is $164 - 130 = 34$ and white extra use is $188 - 130 = 58$. These combine to the pink stripe, so $s = 34 + 58 = 92$.

2. Each leftover is $130 - 92 = 38$, hence total 114.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2011-I-Q01] | Part A (Q1-Q5)

Answer: 085

1. The combined final volume is ten liters with five liters of acid. Conservation gives $4(0.45) + 5(0.48) + k/100 = 5$, so $k = 80$.

2. If x liters go to A, $1.8 + 0.8x = 0.5(4 + x)$, giving $x = 2/3$. The remaining $1/3$ also makes B fifty-percent.

3. Hence $k + m + n = 80 + 2 + 3 = 85$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2011-II-Q01] | Part A (Q1-Q5)

Answer: 037

1. Normalize the actual purchase to one and let consumption be r. Then $1/2 - 2r = (2/9)(1 - r)$.

2. Multiplication by eighteen gives $9 - 36r = 4 - 4r$, so $r = 5/32$. It lies below $1/4$, making the hypothetical leftover nonnegative. The answer is 37.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2012-I-Q04] | Part A (Q1-Q5)

Answer: 279

1. Walking one mile takes the faster traveler fifteen minutes and the slower twenty-four; riding takes ten. If the faster traveler walks x of the first n miles, the slower rides those miles, so equality of arrival times is $15x + 10(n - x) = 10x + 24(n - x)$, or $19x = 14n$. Thus n must be a multiple of nineteen.

2. To verify attainability under the stated handoff rule, let Δ be faster minus slower arrival time at successive posts. When $\Delta < 0$, the slower traveler rides and Δ increases by five; when $\Delta > 0$, the faster rides and it decreases by fourteen. Starting at zero, the first mile gives five.

3. This keeps Δ in the integer interval $[-13, 5]$ and gives $\Delta \equiv 5n \pmod{19}$. Hence it first returns to zero exactly at post nineteen. Then $x = 14$.

4. The common time is $15(14) + 10(5) = 260$ minutes, so $n + t = 279$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2012-II-Q04] | Part A (Q1-Q5)

Answer: 061

1. Let those dimensions be L, W, x. Equal times give $(2L + W - x)/8.6 = (W + x)/6.2 = \sqrt{W^2 + x^2}/5$. The first equality gives $31L = 6W + 37x$.

2. The second equality, squared, is $168W^2 - 625Wx + 168x^2 = (24W - 7x)(7W - 24x) = 0$.

3. If $x = 7W/24$, then $L = 13W/24 < W$, impossible. Thus $x = 24W/7, L = 30W/7$, which also has $0 < x < L$.

4. The ratio is $30 : 7 : 24$, giving 61.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2013-I-Q01] | Part A (Q1-Q5)

Answer: 150

1. For swimming speed s, the three times are $0.5/s, 3/s, 1.6/s$ hours. Their sum is $5.1/s = 4.25$, so $s = 1.2$.

2. Cycling speed is twelve mph, and thirty miles take 2.5 hours, or 150 minutes.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2013-I-Q03] | Part A (Q1-Q5)

Answer: 018

1. The two square sides are x, y, so $x^2 + y^2 = (9/10)(x + y)^2$.

2. Rearranging gives $x^2 + y^2 = 18xy$. Dividing by the positive product yields the required value 18.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2013-II-Q01] | Part A (Q1-Q5)

Answer: 275

1. The old time is 396 of the 1440 daily minutes. Multiplying the fraction of a day by 1000 gives $396 \cdot 1000/1440 = 275$ decimal minutes.

2. The display is $2{:}75$, and the required value is 275.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2014-I-Q04] | Part A (Q1-Q5)

Answer: 049

1. Let east and west train speeds be u, v and length L. The one-minute conditions give $u - 20 = v + 20 = 60L$, hence $u = 60L + 20, v = 60L - 20$.

2. Overtaking the westbound rider requires $v > 20$. The tenfold time ratio gives $u + 20 = 10(v - 20)$, so $60L + 40 = 10(60L - 40)$.

3. Thus $L = 22/27$ miles, giving 49. The derived westbound speed $260/9$ exceeds twenty, verifying the branch.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2014-II-Q01] | Part A (Q1-Q5)

Answer: 334

1. The per-minute rates are $1/900, 1/600, 1/450$. After ninety minutes, one-tenth of the room is done.

2. The next two-fifths take $360(2/5) = 144$ minutes at combined rate $1/360$. The final half takes one hundred minutes at rate $1/200$.

3. The total is $90 + 144 + 100 = 334$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2017-II-Q05] | Part A (Q1-Q5)

Answer: 791

1. Let the four-number total be s. The six pair sums split into three complementary pairs, each totaling s. If x, y formed one pair, the four given values would split into two equal-sum pairs; none of the three pairings does so ($509 \neq 521$, $476 \neq 554$, $423 \neq 607$).

2. Therefore two given values must themselves be a complementary pair. Since all six sums total $3s$, $x + y = 3s - 1030 \leq 3(320 + 287) - 1030 = 791$.

3. Attainment is given by the four distinct numbers $103/2, 275/2, 365/2, 471/2$. Their remaining pair sums are373 and418, totaling791.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2018-II-Q01] | Part A (Q1-Q5)

Answer: 800

1. Let $AB = x$. Paul covers four-fifths of the initial gap $1800 - x$ before meeting Eve, then retraces this distance. His total is $8(1800 - x)/5$.

2. In the same time Ina covers x, while Paul runs twice as fast, so $8(1800 - x)/5 = 2x$. Solving gives $x = 800$ m.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2022-I-Q05] | Part A (Q1-Q5)

Answer: 550

1. Place the slower start at $(0,0)$ and faster start at $(D,0)$. The target is $(D/2,264)$. For common time t, write $u=D/(2t)$ and $v=264/t$. The water-relative velocity vectors are $(u-14,v)$ and $(-u-14,v)$.

2. Subtract their squared speeds: $(u+14)^2-(u-14)^2=80^2-60^2$, so $56u=2800$ and $u=50$. Then $v^2=60^2-36^2=2304$, giving $v=48$.

3. Thus $t=264/48=11/2$ minutes, and $D=2ut=550$ m.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2023-II-Q04] | Part A (Q1-Q5)

Answer: 273

1. Subtract the first two equations to obtain $(y-4)(x-z)=0$. Thus either $y=4$ or $x=z$.

2. If $y=4$, then $x+z=15$ and $xz=44$, giving $x=4$ or 11.

3. If $x=z$, then $x(y+4)=60$, so $x\neq 0$. Substitution into $x^2+4y=60$ gives $x^3-76x+240=(x-4)(x-6)(x+10)=0$. Every resulting value has a valid real companion $y=60/x-4$.

4. Therefore $S=\{-10,4,6,11\}$ and the square sum is $100+16+36+121=273$.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2024-I-Q01] | Part A (Q1-Q5)

Answer: 204

1. Let the stop last u hours. Then $9/s + u = 4$ and $9/(s+2) + u = 12/5$.

2. Eliminating u gives $18/[s(s+2)] = 8/5$, so $s = 5/2$ (the other root is negative). Thus $u = 2/5$ h.

3. The requested speed is 3 km/h. The total is $60(9/3 + 2/5) = 204$ minutes.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2026-I-Q01] | Part A (Q1-Q5)

Answer: 277

1. Let first speed be v and its travel time $T > 2$. Equal distances give $vT = (v+2)(T-1) = (v+9)(T-2)$.

2. The first equality gives $v = 2T - 2$; the last gives $2v = 9T - 18$. Solving yields $T = 14/5, v = 18/5$, so distance $252/25$ and the answer 277.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2000-I-Q07] | Part B (Q6-Q10)

Answer: 005

1. Let $t = z + 1/y$. Because $xyz = 1$, expanding the product of the three cyclic expressions gives $(x + 1/z)(y + 1/x)(z + 1/y) = 2 + (x + 1/z) + (y + 1/x) + (z + 1/y)$.

2. Thus $145t = 36 + t$, yielding $t = 1/4$ and the requested sum 5.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2000-I-Q10] | Part B (Q6-Q10)

Answer: 173

1. Let S be the total sum. Each equation says $2x_k = S - k$. Summing gives $2S = 100S - 5050$, hence $S = 2525/49$.

2. Therefore $x_{50} = (S - 50)/2 = 75/98$, and the requested sum is 173.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2010-I-Q09] | Part B (Q6-Q10)

Answer: 158

1. Put $t = xyz$. Multiplying $x^3 = t + 2, y^3 = t + 6, z^3 = t + 20$ gives $t^3 = (t + 2)(t + 6)(t + 20)$.

2. After cancellation, $7t^2 + 43t + 60 = 0$, whose roots are $-4, -15/7$. Both are achieved by the real cube roots of the three expressions: their product has cube t^3, hence equals t.

3. The desired sum is $28 + 3t$, maximized at $t = -15/7$, giving $151/7$ and answer 158.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2011-I-Q06] | Part B (Q6-Q10)

Answer: 011

1. Vertex form gives $y = a(x - 1/4)^2 - 9/8$. At $x = 1$, its integer value j is $9(a - 2)/16$.

2. Thus $a = 2 + 16j/9 > 0$, requiring integer $j \geq -1$. The least choice is $j = -1$, giving $a = 2/9$ and answer 11.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2021-II-Q07] | Part B (Q6-Q10)

Answer: 145

1. From the first two equations, $ab = 3c - 4$. Grouping the cubic relation gives $ab(c + d) - 3cd = 14$, hence $d = (3c^2 - 4c - 14)/4$.

2. Substitute in $abcd = 30$ to get $(c + 2)(3c - 10)(3c^2 - 4c + 6) = 0$. The last factor has no real root. For $c = 10/3$, $ab = 6$ and $a + b = -3$ would give a negative discriminant, so this case is impossible.

3. Thus $c = -2, d = 3/2, ab = -10$. The square sum is $9 - 2(-10) + 4 + 9/4 = 141/4$, giving 145.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2001-I-Q11] | Part C (Q11-Q15)

Answer: 149

1. If the selected columns are c_i, old and new numbers are $(i-1)N + c_i$ and $5c_i - 5 + i$. The first two equations yield $c_1 = 5c_2 - 3$ and $N = 24c_2 - 19$.

2. The remaining three are $2N + c_3 = 5c_4 - 1$, $3N + c_4 = 5c_5$, $4N + c_5 = 5c_3 - 2$. Eliminating gives $124c_5 = 89N + 7$.

3. Substituting for N requires $7c_2 \equiv 18 \pmod{31}$, so $c_2 \equiv 7 \pmod{31}$. The smallest positive choice gives $N = 149$. Columns $(32, 7, 141, 88, 107)$ lie in range and verify every equation, proving attainability.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2014-I-Q14] | Part C (Q11-Q15)

Answer: 263

1. Add one to each fraction and four to the right side, obtaining $x\sum_{k\in\{3,5,17,19\}}1/(x-k)=x(x-11)$. The candidate $x=0$ is valid; otherwise divide by x and set $t=x-11$.

2. Pair symmetric denominators: $2t/(t^2-64)+2t/(t^2-36)=t$. Besides valid $t=0$, divide by t and set $u=t^2$.

3. The resulting equation is $u^2-104u+2504=0$, giving $u=52\pm\sqrt{200}$. Neither equals the excluded pole squares $36,64$.

4. The largest candidate is $11+\sqrt{52+\sqrt{200}}$, exceeding all other sign choices and $x=0,11$. Hence the sum is 263.

Source / 来源

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2015-II-Q14] | Part C (Q11-Q15)

Answer: 089

1. Put $s = x + y, p = xy$. Factoring gives $p^4 s = 810$ and $p^3(s^3 - 3ps) = 945$. Both s, p are nonzero.

2. Divide the second equation by the first: $(s^2 - 3p)/p = 7/6$, so $s^2 = 25p/6$ and $p > 0$. Squaring $p^4 s = 810$ then gives $(25/6)p^9 = 810^2$, hence $p^3 = 54$.

3. The second equation gives $x^3 + y^3 = 945/54 = 35/2$. Therefore the target is $2(35/2) + 54 = 89$.

Source / 来源